

Active InferAnt Stream 008.1 ~ Onboarding to Active Inference across back

ActiveInferAntStream | Onboarding to Active Inference across backgrounds & languages | 1:11:08

All right, welcome to Active Inference Stream number 8.1.

It's December 15th, 2024.

I'm going to start with a GitHub push to the GitHub repo.

Active Inference Institute slash start.

While we have, meanwhile, cursor, current version, 0.43, translating, curricula, the background, visualizing, learning paths, having our agenda up.

So thanks for joining.

If you're watching live, definitely write some interesting things in the chat and I'll look to re-enter it into the stream.

All right, here we are.

Symbols, greetings.

Good evening, Jeff.

Again, this is in GitHub repository Active Inference Institute slash start.

The idea here was to build on some prior perplexity and LLM based methods that we've worked on in different projects earlier in the year.

And kind of continue with this all by all theme across domains and backgrounds intersecting with onboarding and curricula for Active Inference.

This is more than just a translation tool.

It's a framework for understanding and optimizing how knowledge flows through different perspectives.

Modular treatment of topics and perspectives enables flexible application.

Okay.

Okay.

So, first, where are we headed to in terms of the output?

Then, let's look at the scripts that are using perplexity even now to go through it.

So, the written curricula are in English and then they're being translated in chunks to other language.

So, here's a synthetic biologist background introduction curriculum to Active Inference translated into Japanese.

And this is more than 1400 lines long.

That's what it looks like in cursor.

Let's reload the repo.

It's still writing this very large initial push.

We'll look at what the markdowns look like in a few minutes.

That's the translated curriculum.

The written curricula, there's two sets of audiences that we'll trace back to that it's drawing upon.

The first with just their names are individual specific people or different audience profile.

We'll get to how that's specified like a high school student and a middle school student.

So, these are different audiences or different synthetic knowledge domains that are constructed.

We'll, again, look into them like cooking.

So, this is a complete curriculum implicitly in English for somebody in the cooking industry.

Conduct workshops where participants develop new recipes using Active Inference algorithms.

Predicting food safety risks.

Those are the different domains.

So, this pipeline that we're about to go through uses perplexity.ai, large language model, LLM interface, API, to do research on entities, specific or groups of entities,

and different synthetic domains.

We'll look at how they're defined.

Pool those researches.

Those are done in step one.

Step two, write introductions.

Again, going to call perplexity to write out those curricula in chunks in English.

Three, does visualization of a few different kinds on those generated curricula.

Four translates into languages.

Right now, these languages are uncommented out.

There's more languages you can just comment or uncomment.

I will show my perplexity balance at the beginning of the stream,

and then we'll see how much it changes on during the stream.

Okay.

Okay.

\$67.68 in the perplexity bank.

Using the small here on their model card page.

I'm using the LLAMA 3.1 Sonar small, 128 online.

This is the same context length as these more larger parameter models.

Just from short testing though, it's faster and it's cheaper.

So, 67, 68.

Okay.

All right.

So, where we're going is this.

Now, let's look at the visualizations on those curricula from step three.

Then, look at the scripts.

Let's look at the scripts.

Okay.

First, metrics.

So, this is for each of those.

It's just translating chunks in the background.

That can be multi-threaded, but it can also cause more API trouble than it's worth.

Okay.

So, this is for each.

This is a comma separated file.

Open it in LibreOffice.

This is some word count information in the sections and the paragraphs.

Descriptive statistics on curriculum metric.

This file is some summary statistics on those 37 written curricula in terms of the mean and the average and all of that of those descriptive statistics.

Metrics distribution.

Metrics distribution.png has those descriptive document statistics for those 37 curricula.

Okay.

Content analysis.

This is principal component analysis.pca analysis.

On 13 cumulative percent of the variance in the term frequency.

So, it's interesting that the entities are all clustered or more closely clustered over here.

Or at least some of them.

Key concepts.

Top 20 concepts.

Word cloud.

Similarity matrices with hierarchical clustering.

So, here this dendrogram does sort of visually and at the hierarchy monophily topological clustering method confirming this cluster.

For example.

Or there's this relationship.

Logistics with whatever it is.

Spatial web.

Topic distribution.

This looks like the rows are the loading in the different 37 curricula.

The topics are top seven topics and their presentation across.

Interesting that Chiaca works in fashion.

And so, there's a connection with the fabric.

We can see if that pops up again.

Different topic.

Okay.

And topic summaries.

Topic one.

So, this is the one that has this loading across these different fields.

Okay.

Then there's concept maps and learning paths.

Here's the concept maps.

These are just starting points.

All of these are just starting point draft outputs.

But they can be updated from the underlying information.

So, here's for Chris Fields.

For the curricula generated for Chris Fields.

This is a eigenvector centrality network topic graph.

In different communities.

High school student concept map.

Let's see this one.

Free energy principle.

Looks like it's in this one orange community.

Variational, minimize, inference, active, energy.

There's the blue.

Neural, biology, decision, organismal, tool making.

Maybe these have use.

Maybe not.

Middle school student.

But cool.

Nonetheless.

That's concept map.

And here are learning paths.

These are just.

Okay.

We'll see how this one's.

Looks like a very.

Example applications are central.

Connected.

This hub and spoke way to the.

These.

High school student path.

Hmm.

Somebody with a background in ATM.

Transactions.

AI powered ATMs.

Case study two.

And this is for.

Now let's see the video game.

Game design.

Game mechanics.

Narrative design.

User experience.

Agile.

Programming.

Game.

Integrating active inference into game development.

Dynamic difficulty adjustment.

Mental health.

Same example used in the high school case.

Here's the curriculum content.

So it's about.

It's two.

And it's separate in two.

There's the.

Domain analysis.

And then there's the curriculum content.

And then they're just concatenated to get that markdown.

So here's the curriculum itself.

So let's see.

That has.

Welcome message.

Relevance.

Value proposition.

Connection to existing domain knowledge.

Overview of learning journey.

Success stories and examples.

Conceptual foundations.

Core active inference concepts using domain analogies.

Mathematical principles with domain relevant examples.

Practical applications and domain context.

Integration with existing domain frameworks.

Case studies in the domain.

Dynamic difficulty adjustment.

Narrative generation.

For game.

Interactive examples and exercises.

Some of these interactive ones are TBD.

But at least they're scoped there.

Technical framework.

Mathematical formalization using domain notation.

That's kind of interesting.

Maybe could use the letter set or domain of some.

Given config.

Computational aspects with domain tools.

Utilize game engines such as Unreal Engine or Unity.

For integrating AI driven mechanics.

Implementation considerations.

Practical applications.

Domain specific use case.

Implementation.

Advanced topics.

Printing computer interface.

Virtual reality.

Augmented reality.

BCI.

VR.

AR.

Research direction.

Collaboration possibility.

Further readings.

Books.

Practical implementation steps.

Addressing.

Books.

Practical implementation steps.

Addressing.

Potential questions.

Conclusion.

References.

Okay.

Let's just briefly look at that for Blake.

And then look at scripts.

Okay.

Okay.

Let's just briefly look at that for Blake.

And then look at scripts.

Okay.

Let's just briefly look at that for Blake.

And then look at scripts.

Okay.

for Blake and then look at scripts.

Alright, to the scripts.

Let's look at how some of the translations look.

New curriculum, Russian synthetic biology.

Okay, these sections are empty.

Repeating.

Repeating.

English.

Back to Russian.

English.

Code.

Russian.

Okay, Hindi language.

Somebody with the RSA cryptographic background.

Okay, repeating.

Not filling out sections.

This could just be a prompt error.

Doing the JSON or nuanced approaches to documents, blocks of docs, these kinds of approaches, that'll add a lot of value.

This is the big blob method.

The all by all folder script method.

Okay, here are the scripts.

Okay, there's two scripts with a beginning prefix one underscore research domain and research entity.

So, there's domain and entity folders.

Each of these script ones will iterate over the markdown files in the synthetic domains.

And within each entity subfolder, just the py.

So, the idea here is whether it's a purely synthetic Python.

I just say generate statements looking like a high school student from chrisfields.py or just fields.py format.

And then those can be versions.

That's the pure synthetic LLM based approach.

There's also generating transcripts from videos.

So, these were like transcripts from chrisfields giving different lectures, different papers, different podcast appearances.

You could have one to many.

It could be all the literature of a researcher.

It could be they could bring in their private notes.

What you could be using a local LLM or none.

Generate these py's that are used.

So, the py's are the pseudo code.

The quote, this is not a py file.

Representations of key attributes, quotes, perspectives.

It can be written in structured lists.

So, basically just a bullet point list.

All the way on through having here for Carl Frist.

I think there's some numerical estimates or some panpsychism point seven.

So, then it can have also these other expressivities with the pseudo code at the semantic layer.

Or these could interface with, they could have some sort of schema to be used in a more structured way as well.

And these are just transcripts and other things that are dropped into a folder.

Markdowns.

And then just within here, given the markdowns in this folder, add comprehensive, accurate statements to this py.

That's what is like the entity specific or the audience specific payload or context is the single py in each of these subfolders.

Whereas for the domains, these domains, there's, for the context here, there's basically two kinds.

There's free energy principle and active inference dot md.

Which is just examples, facts, explanations, definitions, facts, questions about active inference and about other domains.

So, here's the synthetic hymenoptera.

So, ants, bees, wasps, and so forth.

And then, to add on to this, just, I'll use control I to use the agent mode in the current cursor.

So, this is only available to, at this time, I believe the anthropic models.

But it uses specific code functions.

And it can do a few different interesting things.

So, early days for this, but pretty cool.

So, here, add questions, accurate facts, examples, information on the hymenoptera.

So, that can be done from a blank slide, blank text file.

So, it's, or, it can be versioned, like, more, and this could also be in an adjacent format.

Or it could be export from a database.

But it's just having these semantic little nuggets and examples.

So, let's see how it edits synthetic hymenoptera.

Okay, so now it's adding.

Adding sections.

Key questions about hymenoptera.

Scientific facts.

Advanced behaviors in hymenoptera.

Evolutionary adaptations in hymenoptera.

Conservation, environmental impact.

So, human interactions with hymenoptera.

So, that would be included the next time that script one is run.

And when script one is run, it outputs into these domain and audience research files.

The output is, as both a JSON and a markdown.

This request, let's look at the prompt in one.

Says, generate, analyze this domain content and identify aspects of their audience characteristics.

Analyze this domain content to understand the typical industry professionals background knowledge and perspective.

According to these structured responses for that domain.

So, this is the research domain.

The research entity question.

Says, generate a comprehensive analysis for this audience.

And also, it can inject in the FEP factual information itself.

Or, you know, saying factual.

Just in the sense of empirical.

From what can be versioned right here.

So, both of those output these research reports.

So, here's the high school student research report.

This is the background analysis.

So, it's just researching the different characteristics and situation for communicating with the average professional from that domain or that audience about active inference.

Okay, that's one.

Two.

So, here, also go to this CodeViz cool tool.

So, this is the CodeViz code analysis.

Let's do it on two.

Let's do it on two.

I hope it's...

Let's redo it for here.

Yeah.

Help us fully analyze.

Let's...

CodeViz code visualization tool.

Meanwhile...

Okay, it loads in the API key.

Okay.

So, let's look at the graphical flow.

API key is loaded with this function.

File is loaded with this function.

Load file.

Extract sections.

Extracts subsections from these different splitting signals.

Generate section prompt.

Generate section prompt.

Generate section prompt.

Takes that chunk.

Its name and its content.

And the FEP information.

And responds to that section's need given the FEP content.

With these constraints.

Create a thorough version.

So, this concatenates...

This is the Madlib.

The pre-pro prompt.

That concatenates the information.

And gives a format.

Again, there's other ways to do this.

Get perplexity response.

Calls...

The perplexity API.

Again, using the LLAMA 3.1 small.

Sonar small.

Save section.

Saves one section.

Concatenate section.

Puts them back together.

Save complete curriculum.

Saves the concatenated output.

Process research file.

Iterates through the sections.

Here's the main.

So, this is what is going to be...

...the perplexity API.

Now, the perplexity API.

Now, the perplexity API.

the main. So this is what is going to be now visualized in the graph.

Logging setup occurs. Script directory and the llm keys are pointed to. Base directory for future file references is named. Audience in the domain, those two different research directories, the location of the FEP act-inf information and the output direction.

The directories are made that need to be made. So setup logging. Main initializes setup logging.

Let's see if these other methods. So these are just definitions.

Perplexity API

is called. Perplexity API is loaded. That's the next function it's called.

Next. Process the research file.

With these methods. Extract the section.

Parse Markdown content into discrete curriculum sections for processing.

for each section extracted,

generate the section prompt, make the concatenated

query to be sent to Perplexity

that is the

Madlib component that's being iterated over, and the

Acanf FEP factual part, or empirical part,

and the format output component, and the system prompt, but that's the same

for all here. That

pre-pro prompt, or whatever we call it, concatenated

final payload assembled, is sent through the GetPerplexity

response.

That passes then to the save file

operations,

and here are the outputs.

So kind of cool, I'm sure there's other ways to use this really

interestingly. We can copy it in different formats.

There's Mermaid format there.

Okay, so that's two. It goes through the sections of

the personalized curriculum overview, and then seeks to generate that material.

Let's analyze the next

number three.

Four is another Perplexity script.

This one, as alluded to earlier, there's a target languages at the top.

If it's uncommented, it'll be iterated over.

So that's what's happening right now.

It's synthetic astronomy, German language.

So right here, on the synthetic astronomy run.

Then it'll do Russian, Portuguese, Swahili, Tagalog, and then uncomment or comment the ones that you want,

iterate it over.

However, this is some font mapping.

This has analogous Madlib in curriculum content in English.

Maybe for some languages, it would be better to just go direct to the language too.

But here, there could be a repeat.

It could be a curated of one and shifted to another format, content, and language.

So let's look at the codeviz on three.

Okay.

Collects the curricula, pre-processes them,

analyzes their content, generates the descriptive statistic metrics on the curricula, over curricula.

Here's a dependency check for the visualization components.

Here's a decorator for plotting safely.

Plot safely decorator at plot safely.

This is a decorator being applied to plot content analysis.

Plot safely applied to plot learning paths.

Plot concept maps.

Those are the three folders that are shown in the beginning.

Curriculum visualizations.

There's the metrics initial output.

Then there's these three plot safely sections.

Concept mapped.

Content analysis.

Content analysis.

Learning paths.

Looks like it's still processing the three.

Let's do another GitHub push.

Add a few more languages.

Push to GitHub.

6768.

Reloading perplexity.

Seeing how much just this one thread of continuous chugging with the...

So that was a 65 cents USD from having one thread continually run for however long that was.

So a couple dollars an hour for one thread.

So that's a couple dollars for one thread.

And each...

Let's just do it the LLM way.

Copy in some logs.

Throw in chat.

How long does each of these...

CLAWD 3.5 SONNET 2024 1022 coming through.

So between 9 and 23 seconds.

And there are probably better translation models.

Okay.

This one's still processing.

Maybe it'll get there.

If you're watching live though, ask a question.

Otherwise, I'll just sort of review everything.

So to review...

Where this is at.

What can it do?

Well...

We can learn a lot.

By putting in specific or general entities or domains.

It could be...

This conference...

At this exact time...

At this exact time...

.md.

It could be...

Entity...

An organization...

It could be a specific person...

At an organization...

Or a group...

At an organization...

You can add your own...

You can add your own...

Using...

Whatever...

Background supporting files...

Projected onto a PUI...

Synthetic domains...

Entities...

Research them...

The research prompt could be extended...

Or the research output could be...

Used even...

By itself...

It's in the...

Domain research...

In the JSON and the markdown.

Two...

Write introduction...

Splits...

The...

Personalized...

Domain or audience research...

And...

Writes that curriculum...

For them...

Resulting in a...

In a...

In a...

In a...

In a...

In a...

In a...

In a...

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In a....

curricula. So again, here's, let's go to the high school student, written curricula, high school student, versionable curricula and research on bridging to their background or situation. People can continue to update this and add some high schools. That's analyzed in three, visually, and translated in four. So hopefully these are useful starting

points by themselves. They're just files you can download and send to somebody. You can use the GitHub repo slash start to request collaboration access, make a merge and pull all that fun stuff, become a collaborator.

Um, this was kind of, it's all in the languages folder. So we could have some other top level folders. Of course, this is just a new repo. So lots of ways we can go with it. Um, I'll look now to the to-do, but if anyone's watching live, wants to make a comment or request, otherwise I'll go to the to-do, organize a few of those, add a few more things that have come up, see what anyone writes in the chat.

Um.

Um.

Okay.

Okay.

Okay.

Researcher integration.

Implement ORCID API integration.

ORCID and other identity API integrations.

Get research profile information on researchers in open source databases.

Have data interfaces, visual interfaces for people adding information.

Publication scraping.

Of course, we're legal.

Research domain analysis.

This could be using perplexity with internet to look for other context on the fields of somebody's research as scraped and submitted.

It could be some other research.

Design researcher profile structure.

So this is maybe where it could connect with something like within or across organizational platform.

Different user profiles.

Validations.

Okay.

Okay.

So this is developing researcher entity domains researchers.

So that's the first.

Improve our handling and structuring and ingress and accounting for of entity audience information and synthetic domain information.

Database implementation.

Database implementation.

Clarify.

Design and clarify.

Clarify.

Clarify.

Clarify.

Clarify.

Clarify.

Clarify.

Clarify.

Clarify.

Clarify.

Clarify.

Clarify.

Clarify.

Clarify.

so versioning information, knowing which person's information was used for which input to which model, what parameterization, at what time, etc.

Influent database connectors. Not sure what scale some of this query optimization and caching becomes relevant and economical, probably eventually. Data validation, definitely important. Okay, so this has to do with having a better structured, instead of the all by all, blob by blob, folder by folder, inch by inch approach here, having some better structured database, and I guess it goes into it here in phase three, which let's just look at that one. Use fabric or maybe another agent orchestration or kind of computing or task framework, but one of the previous streams used fabric and there may be others to research here, but again, use that database with more of an expressive pattern language, templated, procedural, operational, nuanced approach. Like if we only want to do one language and one background right now, the fastest way to do it is to comment out all the languages except for that one in the file and then remove temporarily everything from the domains and the entities except for the one that you wanted. But that might work to get one done. But if we wanted to have

each of them done in a custom way, then having some more expressive, reproducible, composable, pre-rendering layer for these operational traces, that would be important.

Back to phase two here, as it laid out. Application extensions. Okay. Organizational support.

Organizational configuration system. So different templates, maybe having a single config folder with its own versioning where the branding or the logo or the styles and the document formats for a given organization could all be handled there. So it sort of could be plug and play and separate out the entity background, for example, from the organization config. And this is more like an interactive interface direction, but to build out better some of these analytics and visualizations within and across curricula and then have a static output dashboards or interactive dashboards using some kind of business intelligence tool.

Use catechism framework, templated structure, question and answer format, grant writing system. So this gets more into the

document genre, modular writing, compositional writing with human and LLM and all that happening. Okay, phase three.

That was the pattern languages for more expressive task execution.

So that could be a better model improvement, model improvement, model improvement, more expressive model. So that could be a better,

it could be using the perplexity API with a different call. It could be replacing that with some other analogous LLM.

A colleague has provided me some information on using it, but if somebody wants to help update these methods,

so that they call some other longer context window, LLM or something, then just make the change.

Multimodal support, that could be really interesting. Making and receiving images.

Yeah, parallelization memory management wasn't an issue on this old laptop with cursor because the LLMs were

via perplexity, but maybe at some scale. Interactive live demo. Interactive tutorials. Community features.

Monitoring and maintenance resource usage. Reporting dashboard.

Documentation. Okay, prioritization. And some notes. Yeah, pretty interesting what it comes up with.

Let's improve the documentation a little bit. And then see if anyone writes something interesting in the chat.

And then maybe do some one other random feature and then call it for this one.

So, let's go.

Develop and improve.

Improvement and add.

Concise, new, accurate information directly in best sections.

To this repeat.

Oh.

I thought we were talking about synthetic commonopter. Let's see where it goes with it though.

And then we'll return to the repeat.

Read me.

That was with control I composer. While that one's going.

Chat.

Improve at.

Read me.

Adding new accurate sections.

So even though it's like in the same window.

Composer is a little better.

Okay.

Accept it.

Now make.

Think.

Thoughtful.

Contributive.

Accurate.

Additions to read me.

For how far.

We've gotten.

And.

Only that far.

The chat.

A little bit of back and forth in and out of the code.

Sections.

Composer.

Composer.

I found.

Just doesn't get into those sorts of.

Issues.

Bug finder.

This looks funny.

I think it's.

It uses.

A custom cost.

So it.

One time it estimated it at like \$30.

To do this.

It's like.

There's bugs for free outside.

End of the code.

Okay.

Now.

Updating the read me.

Okay.

Focusing a little bit on the hematoptera.

But.

Okay.

Okay.

Current domain.

So this is like.

Very.

Hymenoptera related.

Not very helpful for the read me.

Reject it.

Start a new composer trace.

Reindex.

Cursor.

Sometimes it.

It does it on its own.

Other times.

Just to double check.

Again.

Hello, Aubrey Walters.

Okay, this Python virtualization is probably not needed, so that's a little bit invented.

Mmm.

Not useful information.

I like the built with heart.

No.

Really just always shows it's just such a fractal ride.

Sometimes it's so good, sometimes it really can be insidious over different timescales.

These parts were accurate.

Yeah.

Yeah.

I hope that people

find the primary artifacts

to be useful

like if you have a friend

who

is of a certain

expertise

and a certain language

then

if it's not already here

add it to one of the scripts

or make the request

so that

you can send

translated curriculum

just send

okay

here's a bunch of different backgrounds in Portuguese

for somebody who wants to use that

so I'll continue to let this one

run

for now

probably doing a multi-threaded

or some other way would be better

or for some of the languages

it seems to be

leaving a significant

number of blanks

so maybe I won't

do all of it

let's just

do one sort of

semi-manual LLM

so here

translate

readme

into spanish

output

in

readme

spanish

ok

ok

ok

ok

the documents

folder

move in the

stream agenda

and do

one

baseball

yep looks pretty cool

hmm

okay so i was just translating it into one language manually

push github

alright

now in baseball

for edification and more

please help write

intricate

imbricated

deep lore
for baseball
american 1920s
memorable moments
describing
the at readme
and all it entails
for
this
given
where it is all
comma all

let's see what format it does and then we'll say
um
continue
continue to make more
crystalline
and
messian
and
accurate deep

specific

minor particulars
minor particulars
for more
baseball

lore
and
specific
reference with the
readme
and
FEP
synthetic

quote

all

script

kitty style

terminal

interface

synthetic intelligence

in the format of

long

quote

who's on first

wow

okay

okay

let's see but

it has some

knowledge graph analysis of a 1920s in baseball

okay

useful

metric

analysis

baseball's quantum leap

batting average

contextualization

home run frequency analysis

pitching effectiveness

metrics

fielding position

evolution

hmm

okay

getting there

okay

okay

okay

okay

okay

okay

okay

okay

okay

ail

dialogue retaining all the gestalt and specifics, only entirely being in the quote, who's on first style of dialogue. All details must be retained and with more linguistic cultural baseball esoterica and actuality.

This dialogue serves as a meta-commentary on how baseball's complexity mirrors the fundamental principles of both.

Okay.

Remove anything that is not dialogue.

Incorporate everything into the who's on first.

Dialectical style.

Retaining all scenarios.

And reference.

Alright, once we get to this all, or I'll just trim this first part.

We'll push it on GitHub.

Read it.

See if anyone has a comment.

End.

I'll distill everything into pure Avid and Costello style dialogue.

Incorporating all the historical, technical, and theoretical content through their exchange.

Thank you, Claude.

Okay.

That looks good.

Documents.

Documents, baseball.

All right.

Let me tell you about the 1920s Yankees, Costello.

Most complex predictive processing system in baseball history.

Fine, but first tell me, who's on first base?

Now you're talking, who's been minimizing free energy on first since the dead ball era?

The dead what era?

Before they put cork in the balls, when who was still optimizing his predictive model?

What are you talking about?

Who's on first?

Exactly.

Been there since 1920.

Right after Ray Chapman's tragic prediction error with Carl Mays.

I don't know anything about Ray Chapman.

No, no.

I don't knows our shortstop.

Working on coupled predictive models with what on second?

What's on second?

Naturally.

Part of Murderer's Row hierarchical Bayesian network.

statistical anomaly that 1927 infield.

Look, all I want to know is the fella's name on first base.

Who?

Been there since Miller Huggins optimized the team's variational free energy state.

Miller who?

No.

Huggins managed who?

Changed the whole generative model of the game.

Revolutionized how tomorrow predicts today's pitching patterns.

What about today?

No.

What's on second?

Today's our Bayesian belief coordinator in the bullpen with tomorrow.

Tomorrow?

Best predictive pitcher in the league.

Learned everything from Walter Johnson's precision weighted curveball processing.

Johnson?

He's pitching?

No.

He's retired.

But his generative model is still influencing tomorrow's pitch selection paradigm.

Then who's pitching today?

No.

Who's maintaining his Markov blanket on first?

Today's updating his beliefs in the bullpen and tomorrow's starting.

Starting what?

The hierarchical processing of batter prediction errors, of course.

Just like how Ruth transformed from pitcher to slugger through active inference.

Ruth?

Now we're getting somewhere.

He's on first?

No.

Who's on first?

Ruth's busy minimizing league-wide surprise with his home run variational distributions.

!

Home run what?

Statistical revolution, my boy.

Change the whole predictive processing of the sport.

Why, before Ruth optimized his generative model.

Never mind, Ruth.

Who's playing first base?

Now you're getting it.

Who's been optimizing first base defense since they standardized the cork-centered baseball's predictive parameters?

I don't care about cork.

What's the guy's name on first?

No.

What's calculating joint probability distributions with seconds with I don't know?

You don't know?

No.

I don't know's integrating radio broadcast feedback with what?

To minimize spectator's surprise.

Radio?

Sure.

Whole new layer of cultural free energy minimization.

Why, when Miller-Huggins first implemented his platooning strategy...

Clutch his head.

Just...

Who's on first base?

Exactly.

Who's maintaining optimal spatial predictions while Gehrig's generative model emerges through pattern recognition?

[illegible]

